minimize

Canonical Correlation Analysis as Reduced Rank Regression in High Dimensions (Appendix)

0.1 A Neuroscience example: the ABIDE Dataset

In this section we present additional performance results and visualization for the Neuroscience dataset.

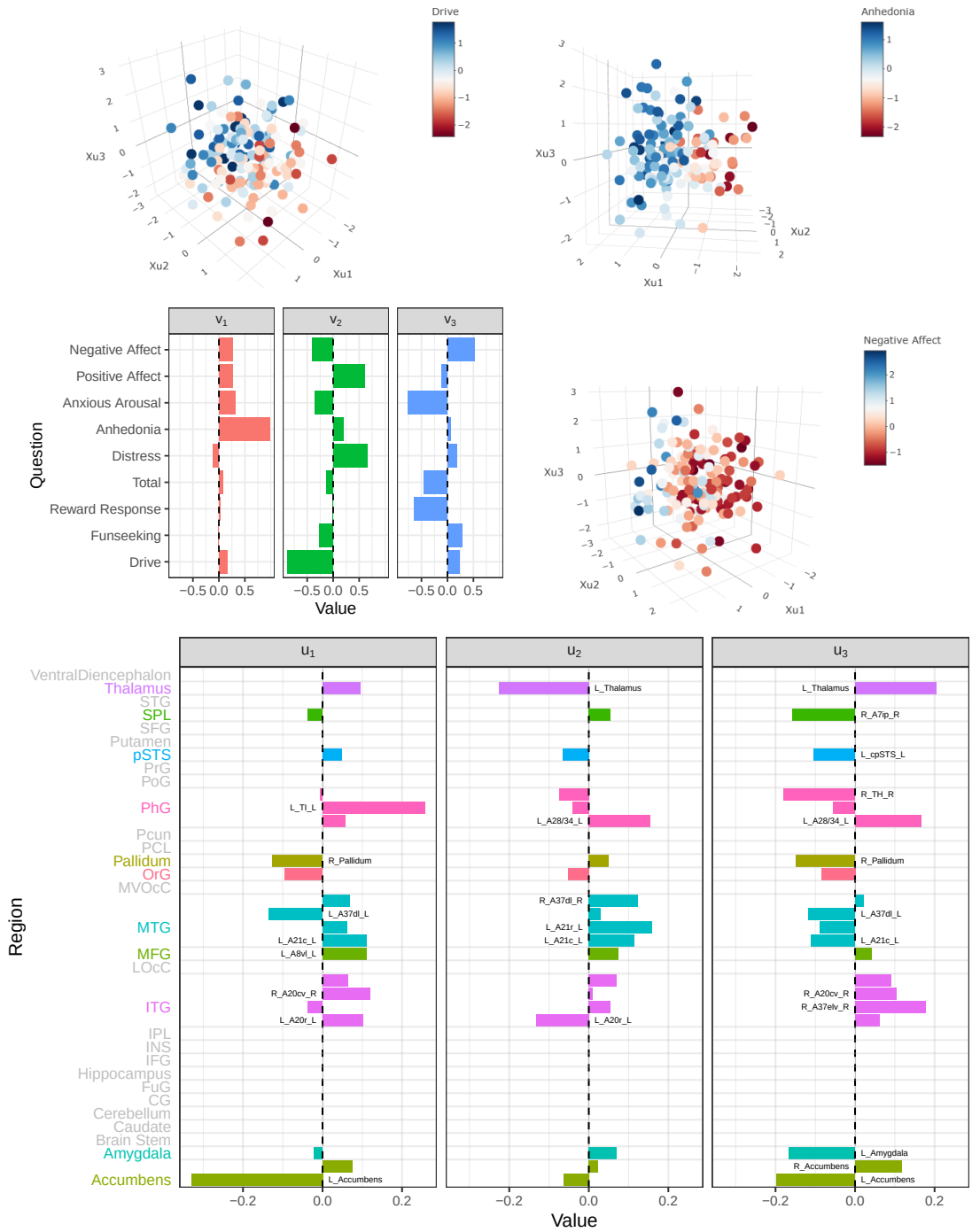

Figure 2: Results on the Neuroscience dataset using group CCAR[3]. Scatter plots: canonical variates $(\boldsymbol{X}\widehat{\boldsymbol{u}}_i)_{i=1}^3$, colored by Drive, Anhedonia and Negative Affect scores. Bar plots: canonical directions $(\widehat{\boldsymbol{v}}_i)_{i=1}^3$ representing the questionnaire and top 20 brain activation directions $(\widehat{\boldsymbol{u}}_i)_{i=1}^3$ grouped along the y-axis and colored according to the brain gyri.

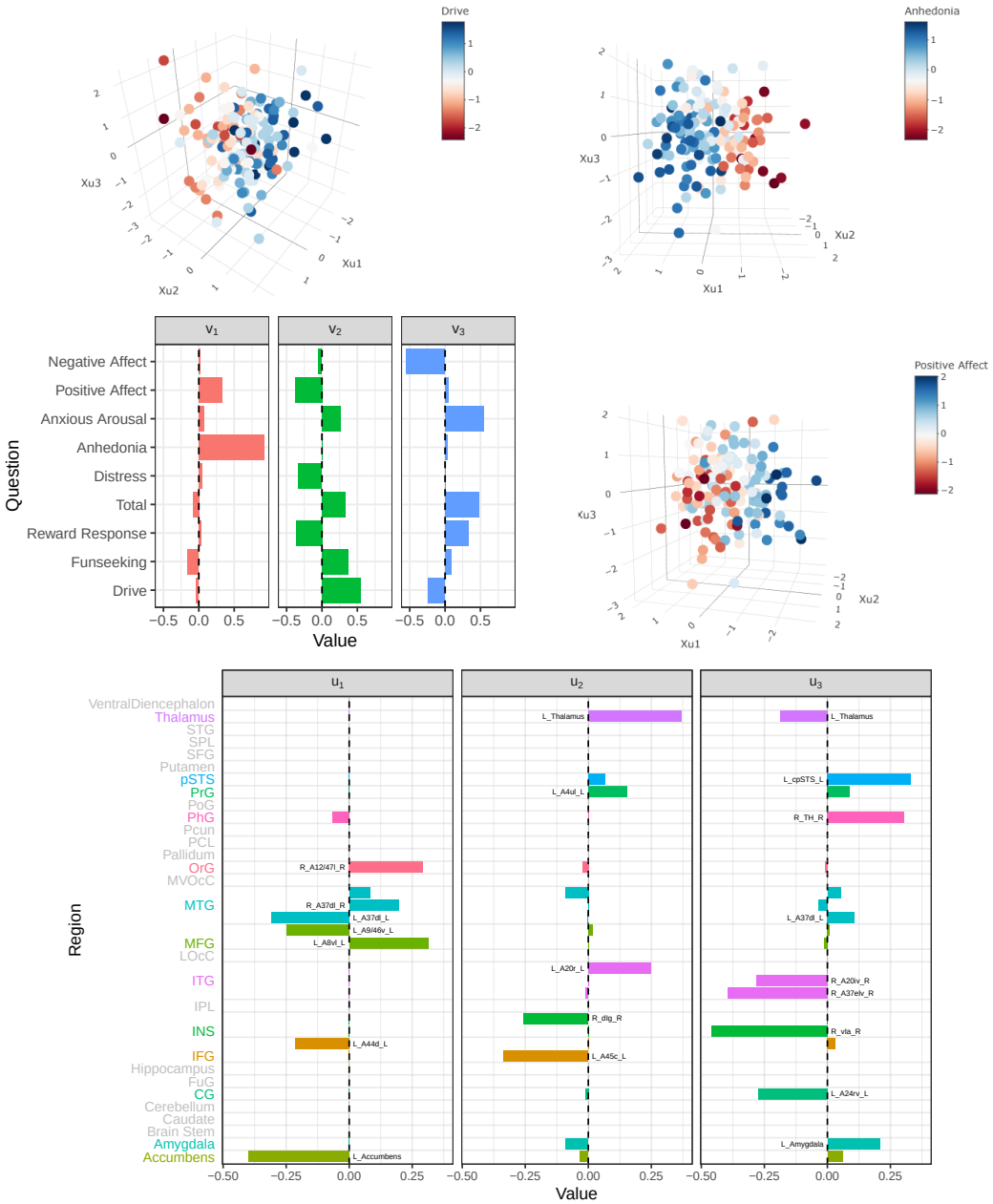

Figure 1: Results on the Neuroscience dataset using SAR. Scatter plots: canonical variates $(\boldsymbol{X}\widehat{\boldsymbol{u}}_i)_{i=1}^3$, colored by Drive, Anhedonia, and Positive Affect scores. Bar plots: canonical directions $(\widehat{\boldsymbol{v}}_i)_{i=1}^3$ representing the questionnaire and top 20 brain activation directions $(\widehat{\boldsymbol{u}}_i)_{i=1}^3$ grouped along the y-axis and colored according to the brain gyri.

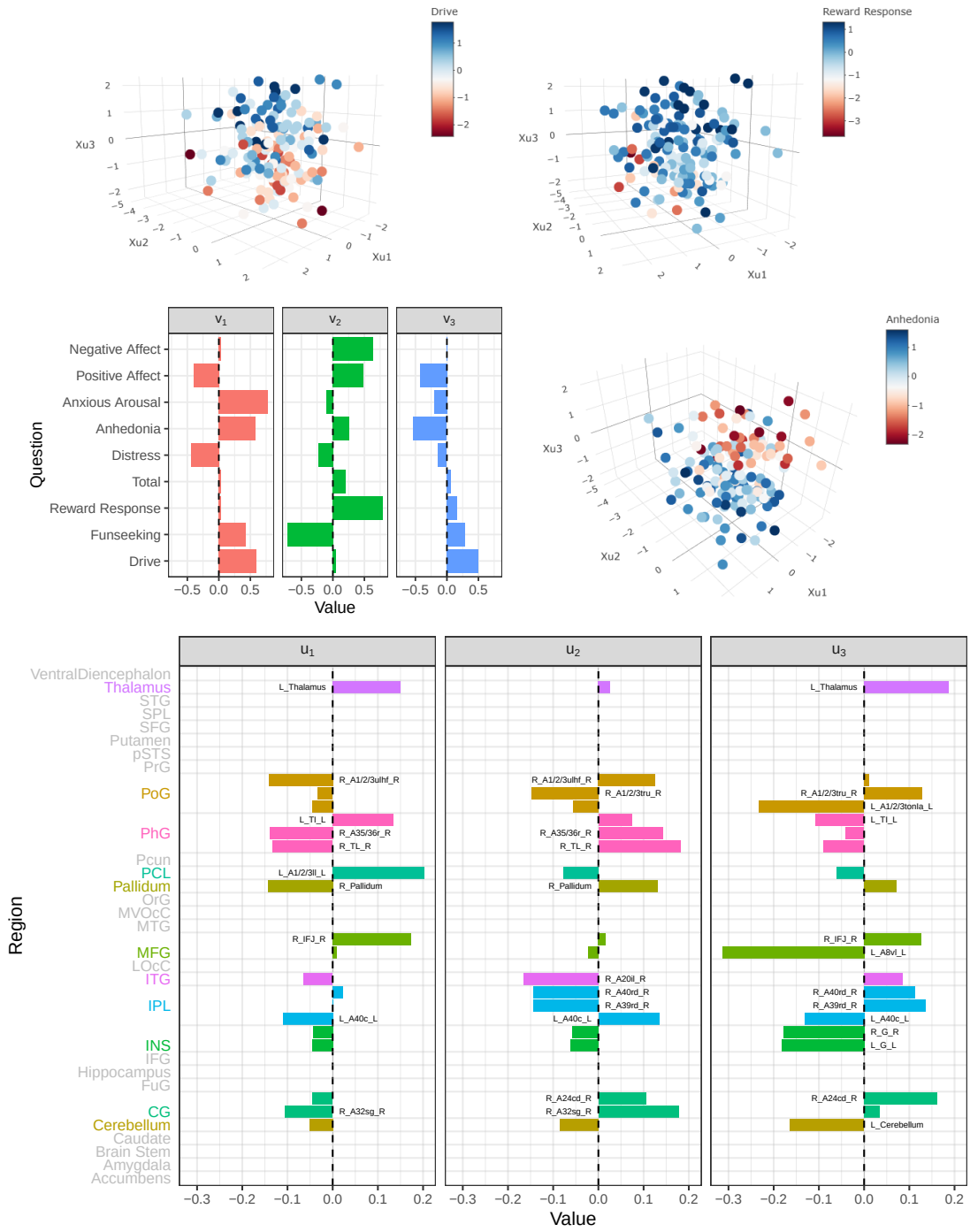

Figure 3: Results on the Neuroscience dataset using graph CCAR^3. Scatter plots: canonical variates $(\boldsymbol{X}\widehat{\boldsymbol{u}}_i)_{i=1}^3$, colored by Drive, Reward Response, Anhedonia scores. Bar plots: canonical directions $(\widehat{\boldsymbol{v}}_i)_{i=1}^3$ representing the questionnaire and top 20 brain activation directions $(\widehat{\boldsymbol{u}}_i)_{i=1}^3$ grouped along the y-axis and colored according to the brain gyri.